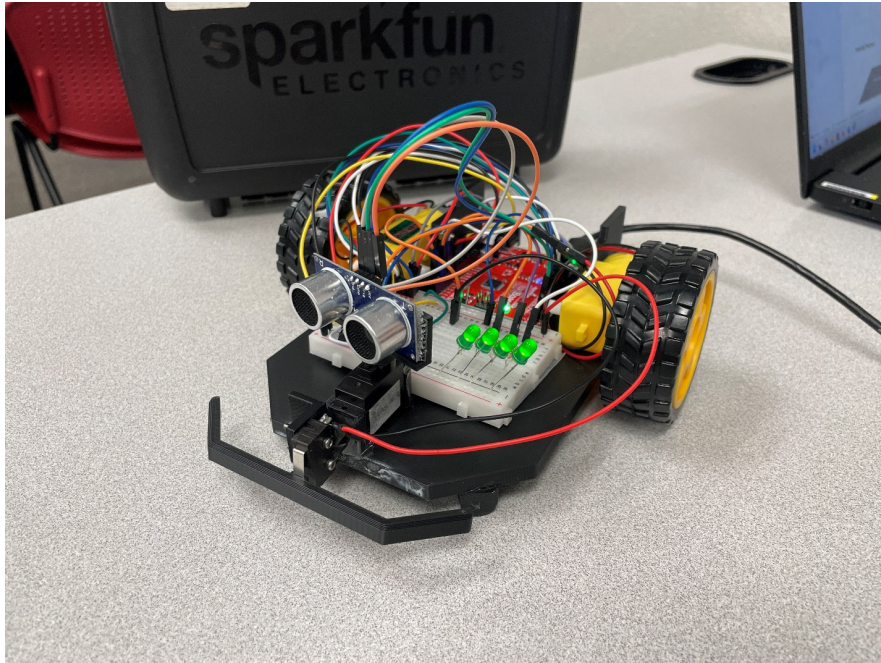


Autonomous Mobile Robot

Coursework, Robotics, Status: Completed

Overview

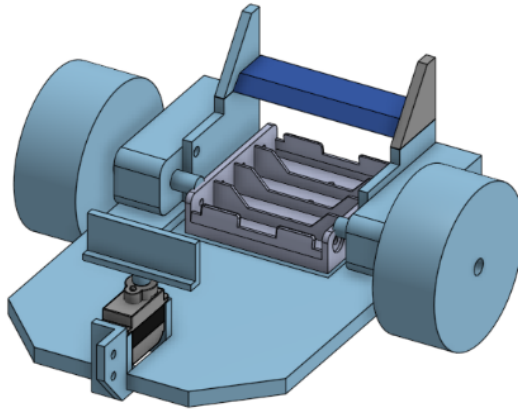
A final project for Computer Applications in Engineering: design and program a robot in MATLAB capable of both driver-controlled and fully autonomous operation, using a custom bump sensor and a sweeping ultrasonic sensor to locate and drive to objects on a field.



The finished robot: chassis, breadboard, ultrasonic sensor, and LED readout.

Chassis and Hardware

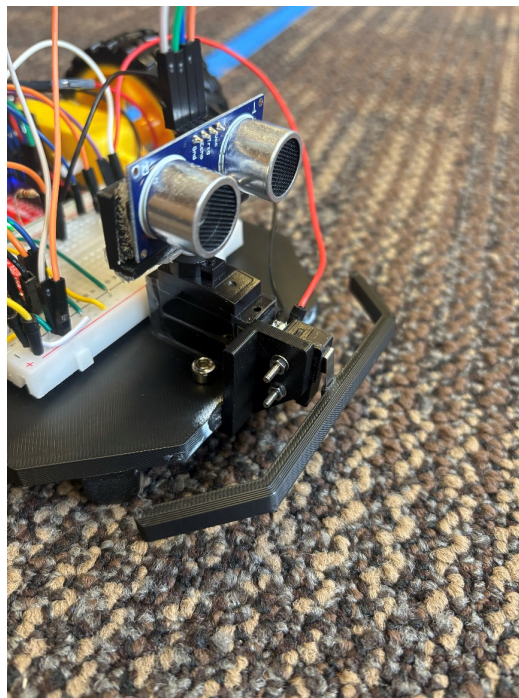
I designed a custom 3D printed chassis in OnShape with room for the Arduino, breadboard, motors, and servo, giving a robust base to prototype different sensor and wiring arrangements. Two drive wheels are paired with small 3D printed skids that keep the chassis level while turning, and four LEDs on the breadboard give a simple visual readout of how many objects have been found.



Chassis CAD assembly, modeled in OnShape.

Bump Sensor: Iterating on a Bad First Design

The first design was a foam and graphite button that acted like a pressure sensor, but contact force varied too much between hits to give a reliable yes or no reading. It was replaced with a 3D printer style limit switch fitted with a wide fender for a larger contact area, which cut down on misses from small turning errors and performed reliably for the rest of the project.



The final bump sensor: a limit switch with a wide 3D printed fender.

Sweeping Ultrasonic Scanning

The first scanning function (`radarScan`) stepped the servo to a fixed angle, then took a reading, one angle at a time. I rewrote it as `radarScanRamp`, which sweeps the servo continuously and samples the ultrasonic sensor during the sweep instead of stopping at each angle, collecting more data in less time. A full 0 to 180 degree sweep gathered 74 distance points in 4.6 seconds, versus 15.8 seconds for the same point count using the original step and read method.

Clump Detection (Object Recognition)

To turn raw scan data into a decision, I wrote `scanAndDetectClump`, which groups nearby distance readings into clumps: points within a set angle and distance of each other. It averages each clump's distance, and returns the angle and distance of the closest one. The servo is then pointed at that angle so the robot can drive toward it. This let the robot target real objects instead of individual noisy readings, in both manual mode (as a plot for the human driver) and autonomous mode (as a heading for the robot to follow).

Operation

- Manual mode: the driver reads the live scan plot and object angle and distance from `scanAndDetectClump`, then drives the robot with a dual motor control function; a bump triggers one LED per object found.
- Autonomous mode: the robot loops through scanning, turning toward the detected clump, driving forward once the object is within a 40 degree cone ahead, stopping on bump contact, lighting an LED, and turning around to search again, repeating until all objects are found or a time limit is reached.

Results

- Top speed: 1 m/s.
- Full scan time: 4.6 seconds (74 points over 0 to 180 degrees).
- Scan resolution: about 20 mm at 0.5 m range; effective range up to 1 m.
- Across three autonomous trials: 9 of 9 objects found, 100% accuracy, with only rare, negligible re-detections of the same object.

What I'd Improve Next

Adding wheel encoders would let the robot orient itself precisely toward a target instead of turning step by step until it's roughly in view, speeding up search and improving driving accuracy. The clump detection logic could also stall if a scan returns no points in range; smarter fallback scanning would make that case more robust.